

TN-2020

Garde 12

Mathematics Part-III

Time: 3.00 Hours.

Max. Marks: 90

Instructions:

- (1) Check the question paper for fairness of printing. If there is any lack of fairness, inform the Hall Supervisor immediately.
- (2) Use Blue or Black ink to write and underline and pencil to draw diagrams.

PART-I

20 × 1 = 20

Note: (i) All questions are compulsory.

(ii) Choose the most appropriate answer from the given four alternative write the option code and the corresponding answer.

1. The value of $\int_0^{\frac{2}{3}} \frac{dx}{\sqrt{4-9x^2}}$ is

(1) π

(2) $\frac{\pi}{6}$

(3) $\frac{\pi}{2}$

(4) $\frac{\pi}{4}$

2. The distance between the planes $x + 2y + 3z + 7 = 0$ and $2x + 4y + 6z + 7 = 0$ is?

(1) $\frac{7}{2\sqrt{2}}$

(2) $\frac{\sqrt{7}}{2\sqrt{2}}$

(3) $\frac{7}{2}$

(4) $\frac{\sqrt{7}}{2}$

3. The position of a particle moving along a horizontal line of any time t is given by $s(t) = 3t^2 - 2t - 8$. The time at which the particle is at rest is :

(1) $t = 3$

(2) $t = 0$

(3) $t = \frac{1}{3}$

(4) $t = 1$

4. If $u(x, y) = e^{x^2 + y^2}$, then $\frac{\partial u}{\partial x}$ is equal to :

(1) $y^2 u$

(2) $e^{x^2 + y^2}$

(3) $2xu$

(4) $x^2 u$

5. The vertex of the parabola $x^2 = 8y - 1$ is:

(1) $\left(0, -\frac{1}{8}\right)$

(2) $\left(-\frac{1}{8}, 0\right)$

(3) $\left(\frac{1}{8}, 0\right)$

(4) $\left(0, \frac{1}{8}\right)$

6. $\rho(A) = \rho([A|B])$, then the system $AX=B$ of linear equations is:

(1) inconsistent

(2) consistent and has a unique solution

(3) consistent

(4) consistent and has infinitely many solution

7. If $(AB)^{-1} = \begin{bmatrix} 12 & -17 \\ -19 & 27 \end{bmatrix}$ and $A^{-1} = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$, then $B^{-1} =$

(1) $\begin{bmatrix} 8 & -5 \\ -3 & 2 \end{bmatrix}$

(2) $\begin{bmatrix} 2 & -5 \\ -3 & 8 \end{bmatrix}$

(3) $\begin{bmatrix} 8 & 5 \\ 3 & 2 \end{bmatrix}$

(4) $\begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$

8. The least possible perimeter (in meter) of a rectangle of area 100 m^2 is:

(1) 50

(2) 10

(3) 20

(4) 40

9. A polynomial equation of degree n always has :

(1) exactly n roots

(2) n distinct roots

(3) n real roots

(4) n imaginary roots

10. $\arg(0)$ is:

(1) ∞

(2) 0

(3) Ω

(4) undefined

11. Subtraction is not a binary operation in:

(1) $\mathbb{Q}$

(2) $\mathbb{R}$

(3) $\mathbb{Z}$

(4) $\mathbb{N}$

12. The radius of the circle $3x^2 + by^2 + 4bx - 6by + b^2 = 0$ is:

(1) $\sqrt{11}$

(2) 1

(3) 3

(4) $\sqrt{10}$

13. The order of the differential equation of all circle with center at (h, k) and radius “a”, where h, k and a are arbitrary constants, is:

(1) 1

(2) 2

(3) 3

(4) 4

14. A random variable X has binomial distribution with $n=25$ and $p=0.8$, then the standard deviation of X is:

(1) 2

(2) 6

(3) 4

(4) 3

15. $\vec{r} = s\hat{i} + t\hat{j}$ is the equation of (s, t are parameters):

(1) zox plane

(2) a straight line joining the points $\hat{i}$ and $\hat{j}$

(3) xoy plane

(4) yoz plane

16. The value of $\sum_{i=1}^{r^3} (i^n + i^{n-1})$ is:

- (1) 0
- (2) $1 + i$
- (3) i
- (4) 1

17. The value of $\int_0^{\pi} \sin^4 x dx$ is:

- (1) $\frac{3\pi}{2}$
- (2) $\frac{3\pi}{10}$
- (3) $\frac{3\pi}{8}$
- (4) $\frac{3\pi}{4}$

18. If $\sin^{-1} x + \sin^{-1} y = \frac{2\pi}{3}$; then $\cos^{-1} x + \cos^{-1} y$ is equal to:

- (1) π
- (2) $\frac{2\pi}{3}$
- (3) $\frac{\pi}{3}$
- (4) $\frac{\pi}{6}$

19. The order and degree of the differential equation $\frac{dx}{dy} + \frac{dy}{dx} = 0$ are:

(1) 2, degree not defined

(2) 1, 2

(3) 2, 1

(4) 2, 2

20. $\tan^{-1}\left(\frac{1}{4}\right) + \tan^{-1}\left(\frac{2}{9}\right) =$

(1) $\tan^{-1}\left(\frac{1}{2}\right)$

(2) $\frac{1}{2} \cos^{-1}\left(\frac{3}{5}\right)$

(3) $\frac{1}{2} \sin^{-1}\left(\frac{3}{5}\right)$

(4) $\frac{1}{2} \tan^{-1}\left(\frac{3}{5}\right)$

PART-II

7 × 2 = 14

Note: (i) Answer any seven question.

(ii) Question number 30 is compulsory

21. prove that $\left(\frac{1+i}{1-i}\right)^3 - \left(\frac{1-i}{1+i}\right)^3$

22. If $(1+i)(1+2i).....(1+ni) = x+iy$, then prove that $2-5-10$
 $..... (1+n^2)=x^2+y^2$

23. Find the value of $\sin^{-1}\left[\sin\left(\frac{5\pi}{4}\right)\right]$

24. Find the magnitude and the direction cosines of the torque about the point $(2, 0, -1)$ of a force $2\hat{i} + \hat{j} - \hat{k}$, Whose line of action passes through the origin
25. Find the value in the interval $\left(\frac{1}{2}, 2\right)$ satisfied by the Rolle's theorem for the function $f(x) = x + \frac{1}{x}, x \in \left[\frac{1}{2}, 2\right]$
26. For the function $f(x) = x^2 + 3x$, calculate the differential df when $x = 2$ and $dx = 0.1$
27. Prove that $\int_0^{\frac{\pi}{2}} \frac{f(\sin x)}{f(\sin x) + f(\cos x)} dx = \frac{\pi}{4}$
28. Find the differential equation of the family of parabolas $y^2 = 4ax$, where 'a' is an arbitrary constant
29. Prove that the identity element is unique if it exists
30. Find the equation of the parabola if the curve is open leftward, vertex is $(2, 1)$ passing through the point $(1, 3)$.

PART-III

7 × 3 = 21

Note: (i) Answer any seven question.

(ii) Question number 40 is compulsory

31. If $A = \begin{bmatrix} 2 & 9 \\ 1 & 7 \end{bmatrix}$ then prove that $(A^T)^{-1} = (A^{-1})^T$
32. If is real, discuss the nature of the root of the equation $4x^2 + 4px + p + 2 = 0$, in terms of P.
33. A concrete bridge is designed as a parabolic arch. The road

over bridge is 40m long and the maximum height of the arch is 15 m. Write the equation of the parabolic arch. Take (0, 0) as the vertex.

34. Find the vector and cartesian equation of a straight line passing through the point (-5, 7, -3) and (13, -5, 2). Find the point where the straight line crosses the xy – plane.

35. If $U = \log(x^3 + y^3 + z^3)$ then find $\frac{\partial U}{\partial x} + \frac{\partial U}{\partial y} + \frac{\partial U}{\partial z}$

36. A random variable X has the following probability mass function:

X	1	2	3	4	5	6
P(X=x)	K	2k	6k	5k	6k	10k

Then find $p(2 < X < 6)$

37. $f(x) = \begin{cases} kx(1-x)^{10}, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$

Find the value of k

38. Prove that $P \rightarrow q \equiv \neg P \vee q$.

39. If the then $\frac{x-x_1}{l_1} = \frac{y-y_1}{m_1} = \frac{z-z_1}{n_1}$ and $\frac{x-x_2}{l_2} = \frac{y-y_2}{m_2} = \frac{z-z_2}{n_2}$ lie on the same plane, then write the number of ways to find the Cartesian equation of the above plane and explain in detail.

40. (a) Test the consistency of the following system of linear equations by rank method

$$x - y + z = -9$$

$$2x - y + z = 4$$

$$3x - y + z = 6$$

$$4x - y + 2z = 7$$

OR

(b) If $2 \cos \alpha < x + \frac{1}{x}$ and $2 \cos \beta = y + \frac{1}{y}$, show that

$$(i) \frac{x^m}{y^n} - \frac{y^n}{x^m} = 2i \sin(m\alpha - n\beta)$$

$$(ii) x^m y^n + \frac{1}{x^m y^n} = 2 \cos(m\alpha + n\beta)$$

PART-IV

7 × 5 = 35

Note: (i) Answer all the questions.

41. (a) Assume that water issuing from the end of a horizontal pipe, 7.5 m above the ground, describes a parabolic path. The vertex of the parabolic path is at the end of the pipe. At a position 2.5 m below the line of the pipe, the flow of water has curved outward 3 m beyond the vertical line through the end of the pipe. How far beyond this vertical line will the water strike the ground?

OR

(b) By vector method, prove that,

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta.$$

42. (a) Find the vector and Cartesian equation of the plane passing through the point (0, 1, -5) and parallel to the straight lines

$$\vec{r} = (\hat{i} + 2\hat{j} - 4\hat{k}) + s(2\hat{i} + 3\hat{j} + 6\hat{k}) \text{ and } \vec{r} = (\hat{i} - 3\hat{j} + 5\hat{k}) + t(\hat{i} + \hat{j} - \hat{k})$$

OR

(b) Evaluate : $\int_{-\pi}^{\pi} \frac{\cos^2 x}{1 + a^x} dx$

43. (a) A police jeep, approaching an orthogonal intersection from the northern direction is chasing a speeding car that has turned and moving straight east. When the jeep is 0.6 km north of the intersection and the car is 0.8 km to the east, the police determine with a radar that the distance between the jeep and the car is increasing at 20 km/hr. If the jeep is moving at 60 km/hr at the instant of measurement what is the speed of the car?

OR

(b) Find the area of the region bounded by x-axis, the curve $y = \cos x$, the lines $x=0$ and $x=\pi$.

44. (a) A square shaped thin material with area 196 sq. units to

make into an open box by cutting small equal squares from the four corners and folding the sides upward.

Prove that the length of the side of a removed square is $\frac{7}{2}$ when the volume of the box is maximum.

OR

(b) If F is the constant force generated by the motor of an

automobile of mass M , its velocity V is given by

$$M \frac{dv}{dt} = F - kv, \text{ where } k \text{ is a constant. Prove that } V = \frac{F}{k} \left(1 - e^{-\frac{kt}{M}} \right)$$

when $t=0$ and $V = 0$

45. (a) In an investigation, a corpse was found by a detective at exactly 8 p.m. Being alert, the detective also measured the body temperature and found it to be 70°F . Two hours later, the detective measured the body temperature again and found it to be 60°F . If the room temperature is 50°F , and assuming that the body temperature of the person before death was 98.6°F , prove that the time of death is 5.26 p.m (5 hrs 26 minutes) (app) $\left[\frac{\log(2.43)}{\log(2)} \approx 1.28 \right]$

- (b) Three fair coins are tossed once. Find the probability mass function, mean and variance for number of heads occurred. Verify the results by binormal distribution.